

Lecture 3: Detection & Segmentation

Segmentation, Localization, Detection

Plan of the lecture:

1. Semantic segmentation
 - Sliding window
 - Fully convolutional
2. Classification & localization
 - Localization as a regression problem
 - Human pose estimation
3. Detection
 - Region proposal
 - Region-based Convolutional Neural Network (R-CNN)
 - Faster R-CNN
4. Instance segmentation
 - Mask R-CNN

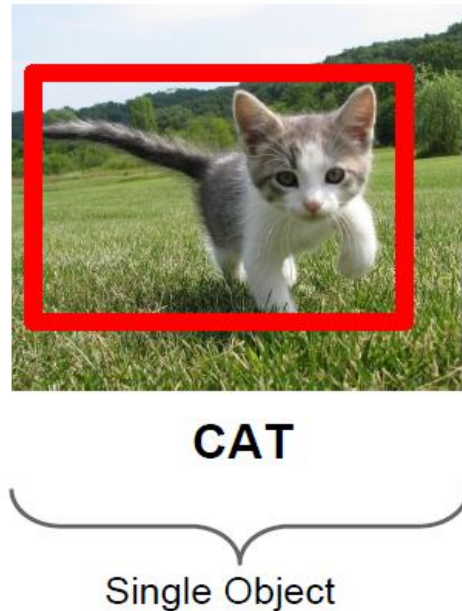
Segmentation, Localization, Detection

Computer Vision Tasks

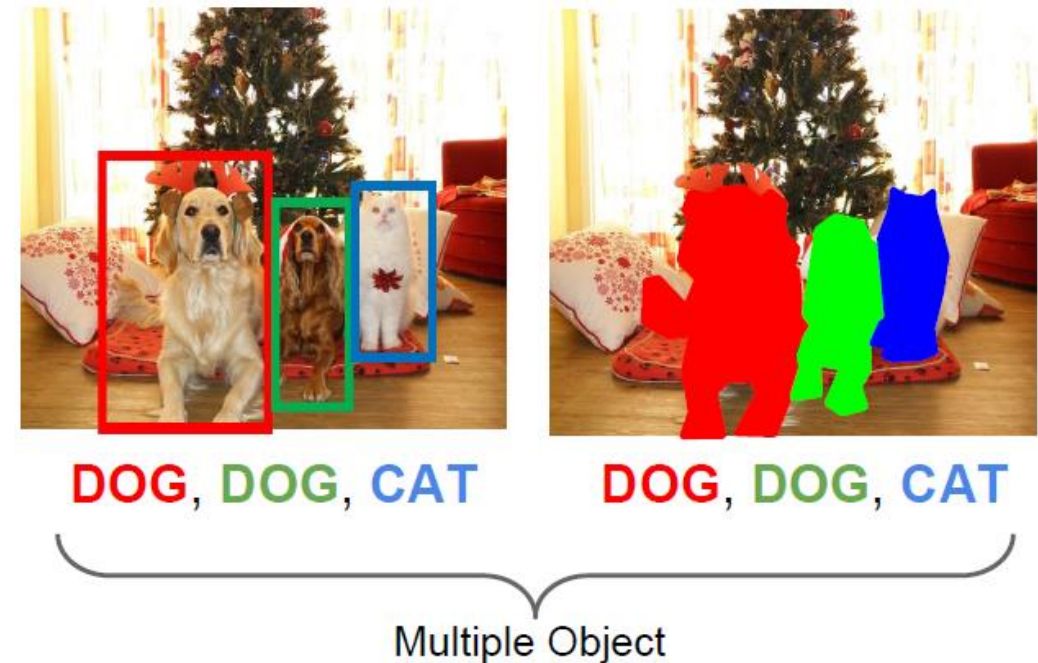
1. Semantic segmentation



2. Classification+Localization



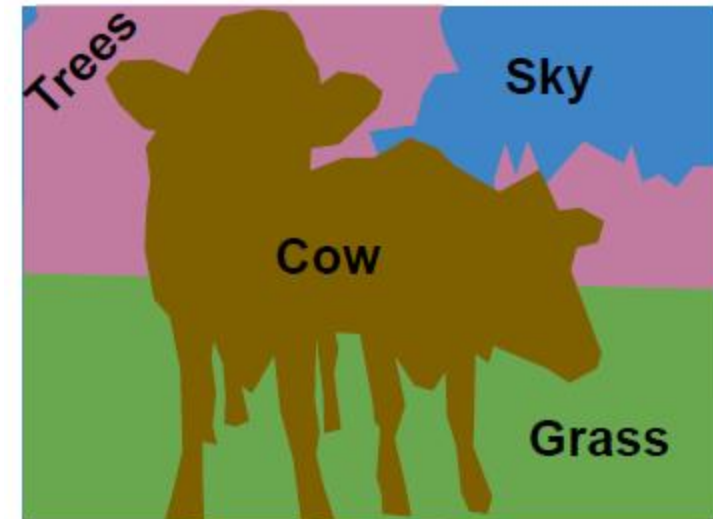
3. Object detection 4. Instance segmentation



1. Semantic Segmentation

□ Main idea

1. Label each pixel in the image with a category label
2. Don't differentiate instances, only care about pixels



1. Semantic Segmentation

□ Main idea

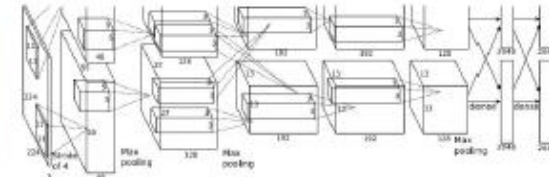
1. How to do semantic segmentation

a. Sliding Window

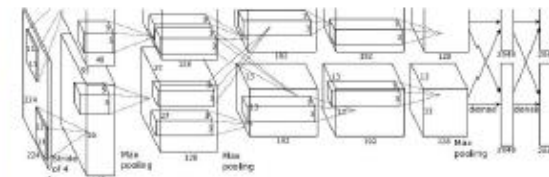
Extract patch



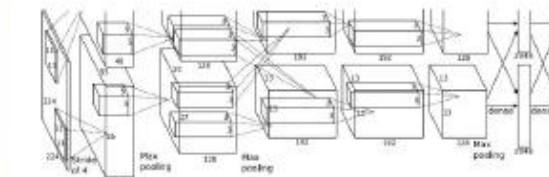
Classify center pixel with CNN



Cow



Cow



Grass

1. Semantic Segmentation

□ Main idea

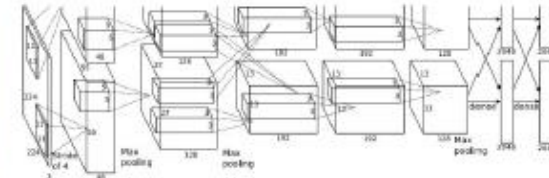
1. How to do semantic segmentation

a. Sliding Window

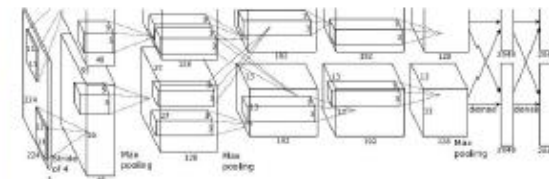
Extract patch



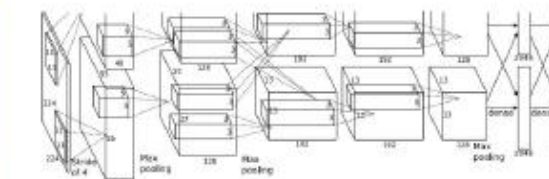
Classify center pixel with CNN



Cow



Cow



Grass

! Very inefficient!
Not reusing shared
features between
overlapping
patches

1. Semantic Segmentation

□ Main idea

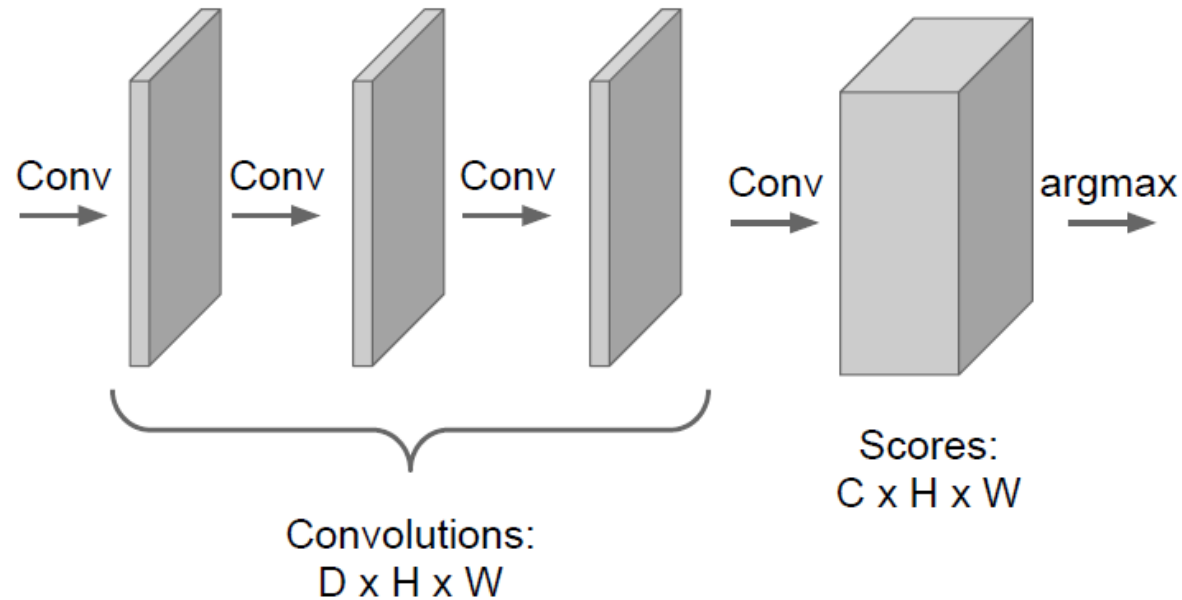
1. How to do semantic segmentation

b. Fully Convolutional

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!



Input:
 $3 \times H \times W$



Predictions:
 $H \times W$

Convolutions at original image resolution will be very expensive

1. Semantic Segmentation

□ Main idea

1. How to do semantic segmentation

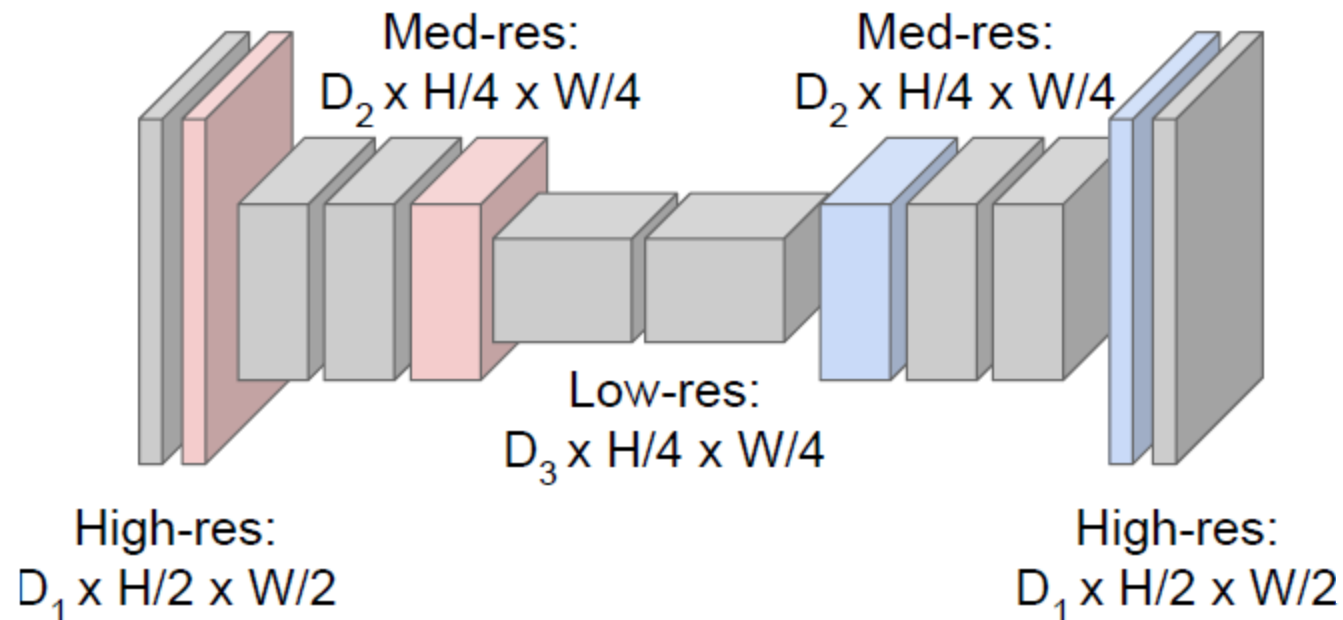
b. Fully Convolutional : smart solution

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Downsampling:
Pooling, strided convolution



Input:
 $3 \times H \times W$



Upsampling:
Unpooling or strided transpose convolution



Predictions:
 $H \times W$

1. Semantic Segmentation

□ CNN In-Network upsampling = Max Unpooling

Max Pooling

Remember which element was max!

1	2	6	3
3	5	2	1
1	2	2	1
7	3	4	8

Input: 4 x 4



5	6
7	8

Output: 2 x 2



...

Rest of the network

1	2
3	4

Input: 2 x 2



0	0	2	0
0	1	0	0
0	0	0	0
3	0	0	4

Output: 4 x 4

Max Unpooling

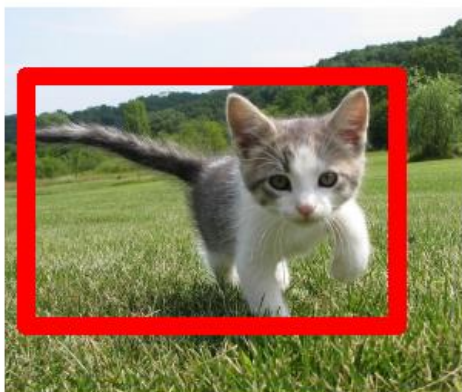
Use positions from pooling layer

2. Classification + Localization

□ Main idea

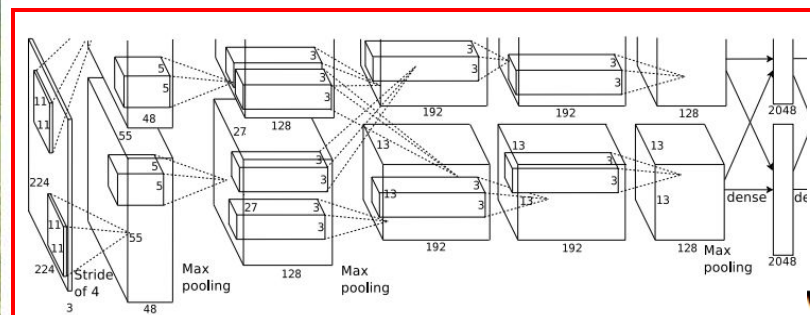
1. How to do Classification & Localization

a. Treat localization as a regression problem

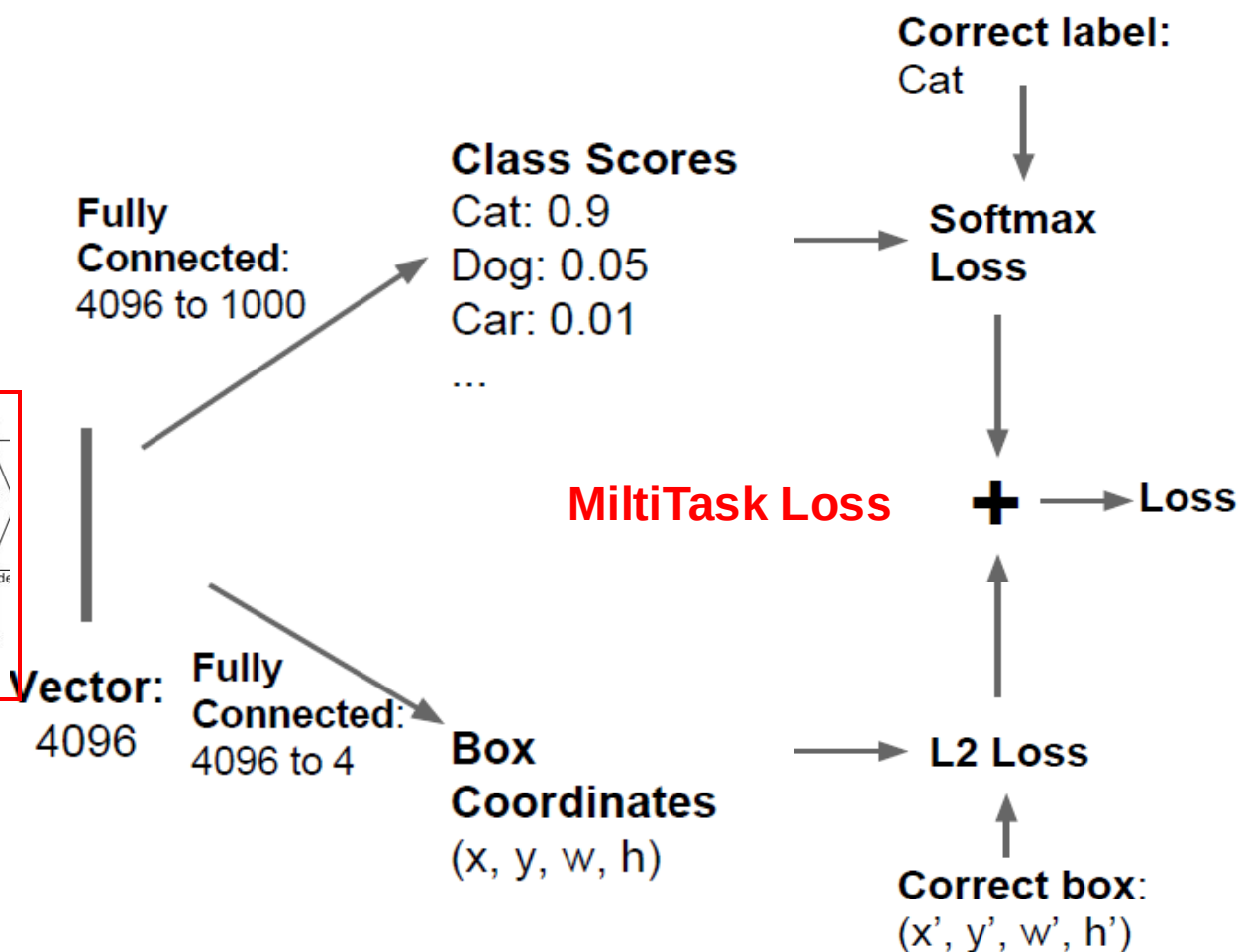


CAT

Single Object



ImageNet (Transfer learning)

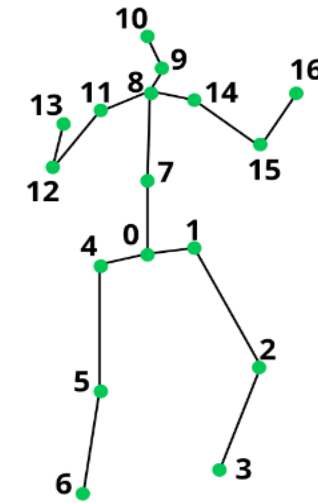


2. Classification + Localization

□ Practical case : Human Pose Estimation



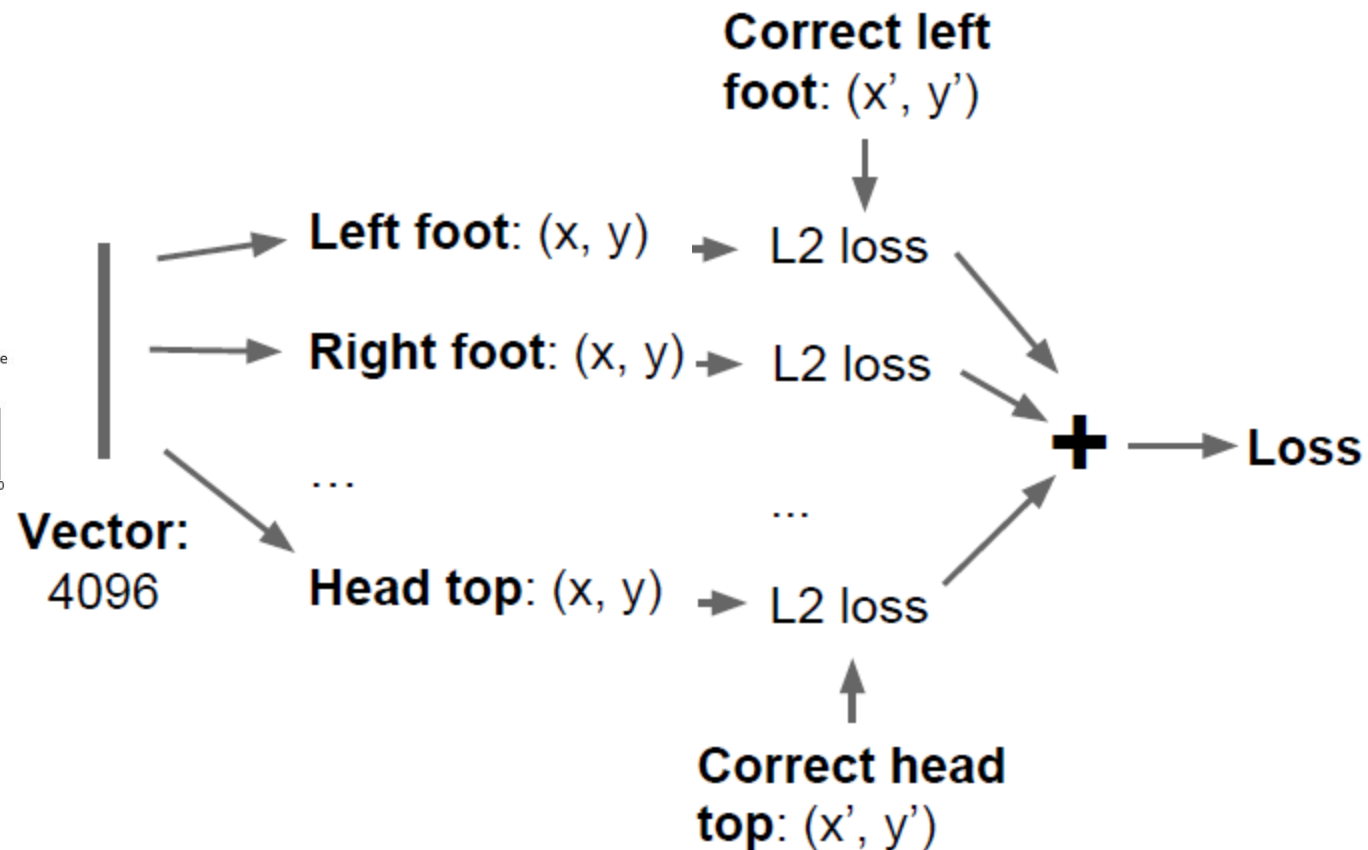
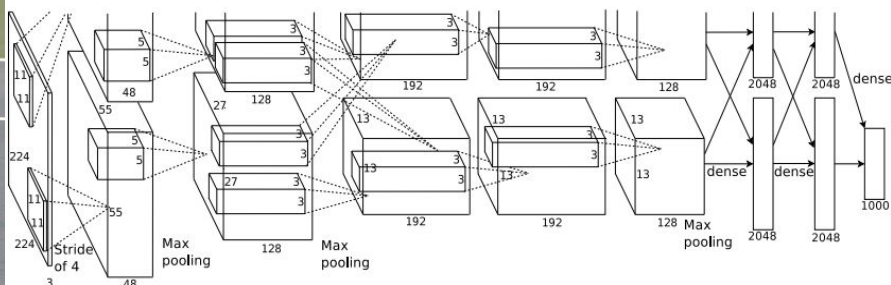
- 0 — Bottom torso
- 1 — Left hip
- 2 — Left knee
- 3 — Left foot
- 4 — Right hip
- 5 — Right knee
- 6 — Right foot
- 7 — Center torso
- 8 — Upper torso



- 9 — Neck base
- 10 — Center head
- 11 — Right shoulder
- 12 — Right elbow
- 13 — Right hand
- 14 — Left shoulder
- 15 — Left elbow
- 16 — Left hand

2. Classification + Localization

□ Practical case : Human Pose Estimation



Toshev and Szegedy, "DeepPose: Human Pose Estimation via Deep Neural Networks", CVPR 2014

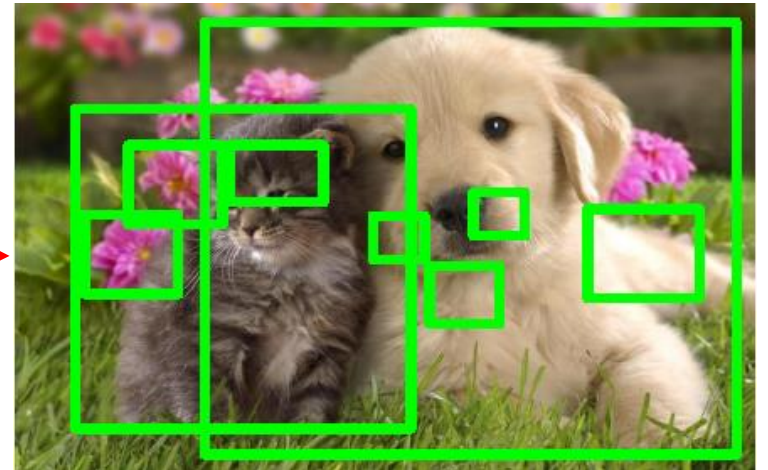
3. Object detection

□ Main idea

1. How to do object detection

a. Region Proposals

- Find “blobby” image regions that are likely to contain objects
- Relatively fast to run; e.g. Selective Search gives 1000 region proposals in a few seconds on CPU

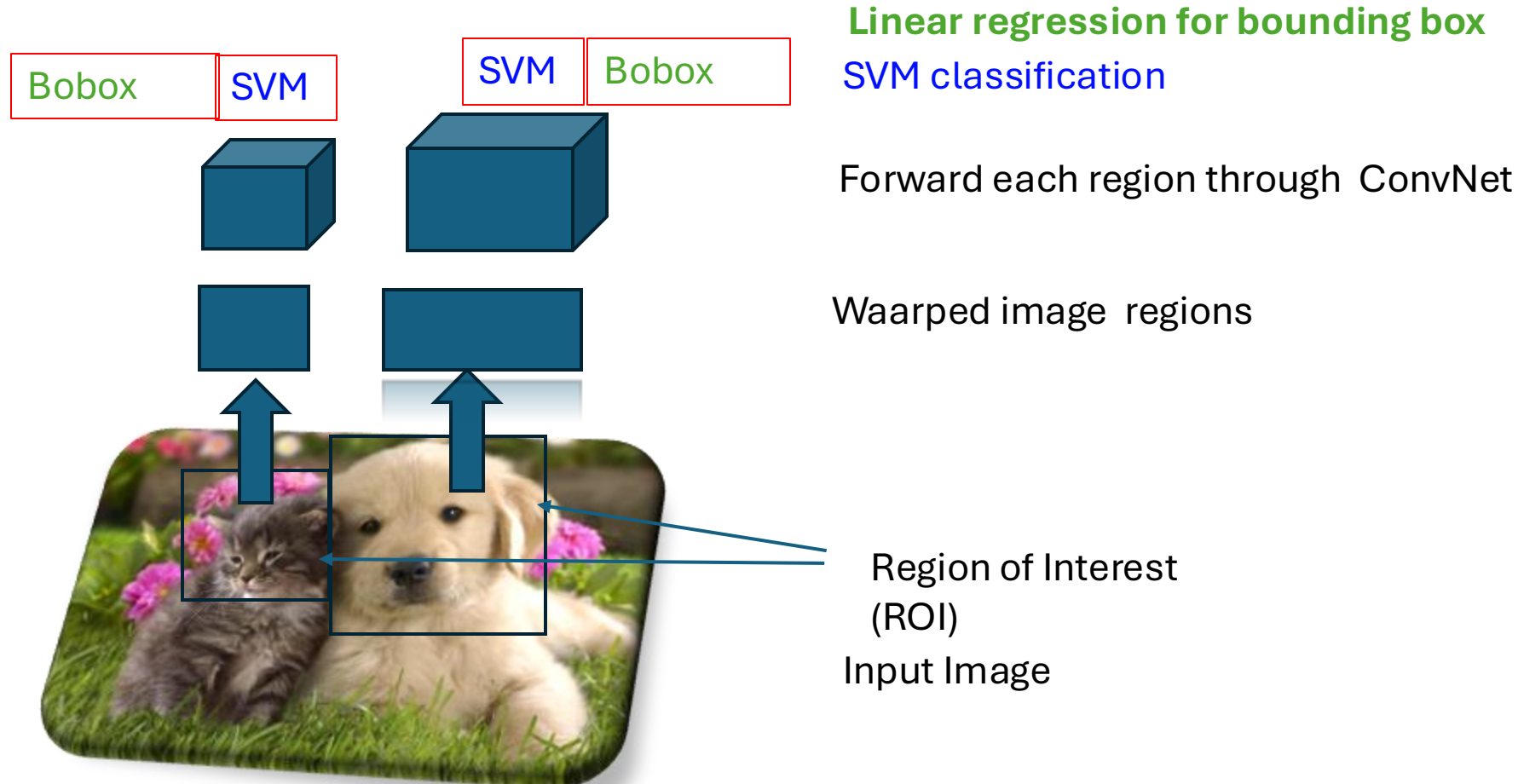


3. Object detection

□ Main idea

1. How to do object detection

a. Region-based Convolutional Neural Network (R-CNN)



Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014.

3. Object detection

❑ R-CNN: Problems

- Ad hoc training objectives
 - Fine-tune network with softmax classifier (log loss)
 - Train post-hoc linear SVMs (hinge loss)
 - Train post-hoc bounding-box regressions (least squares)
- Training is slow (84h), takes a lot of disk space
- Inference (detection) is slow

3.1 Fast R-CNN

□ Main idea

1. How to make it work



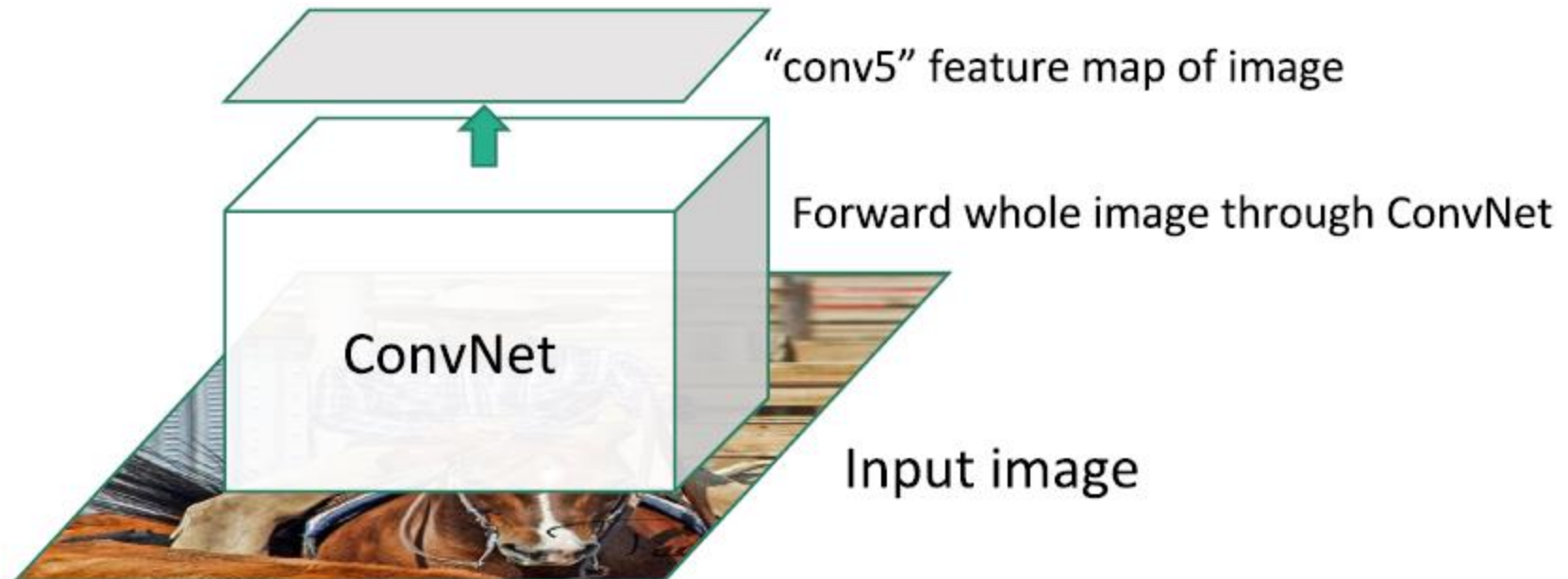
Input image

Girshick, “Fast R-CNN”, ICCV 2015.

3.1 Fast R-CNN

□ Main idea

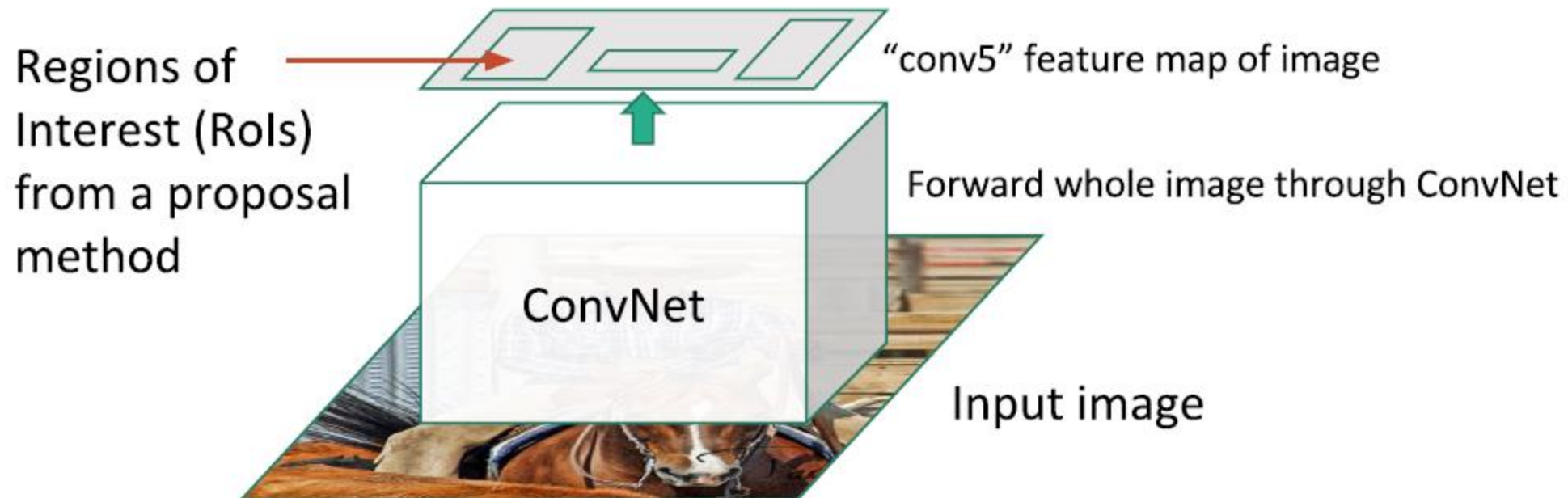
1. How to make it works



Girshick, "Fast R-CNN", ICCV 2015.

3.1 Fast R-CNN

□ Main idea

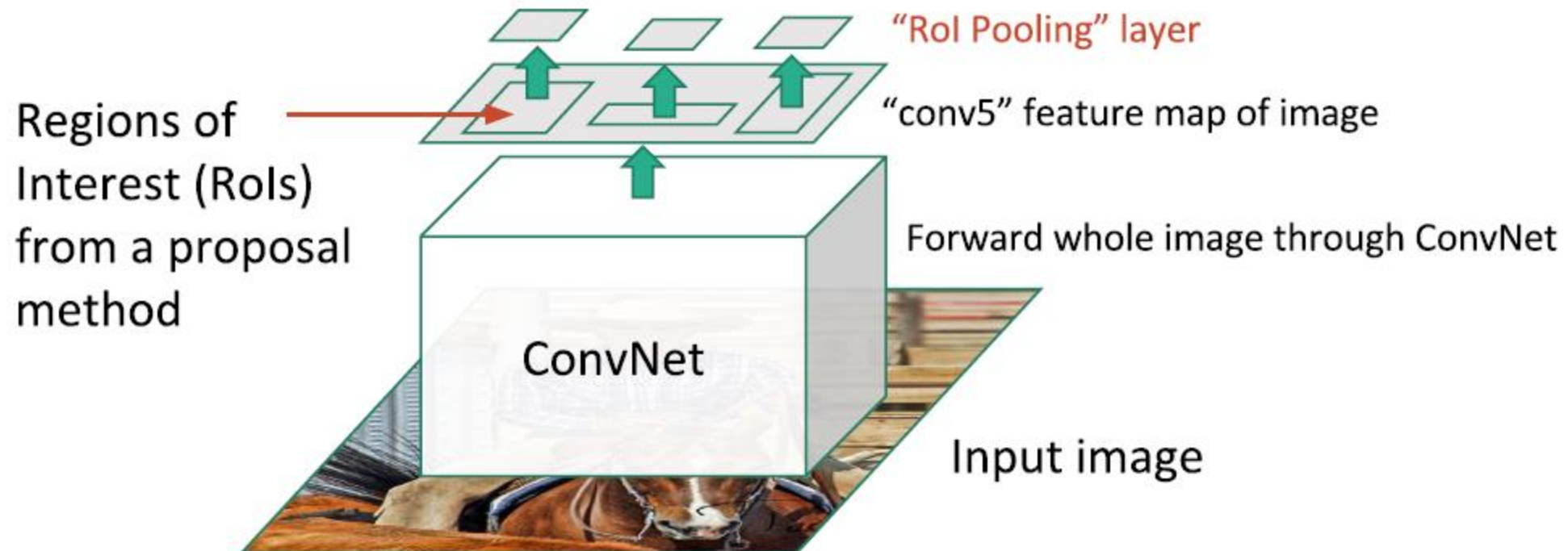


Girshick, "Fast R-CNN", ICCV 2015.

3.1 Fast R-CNN

□ Main idea

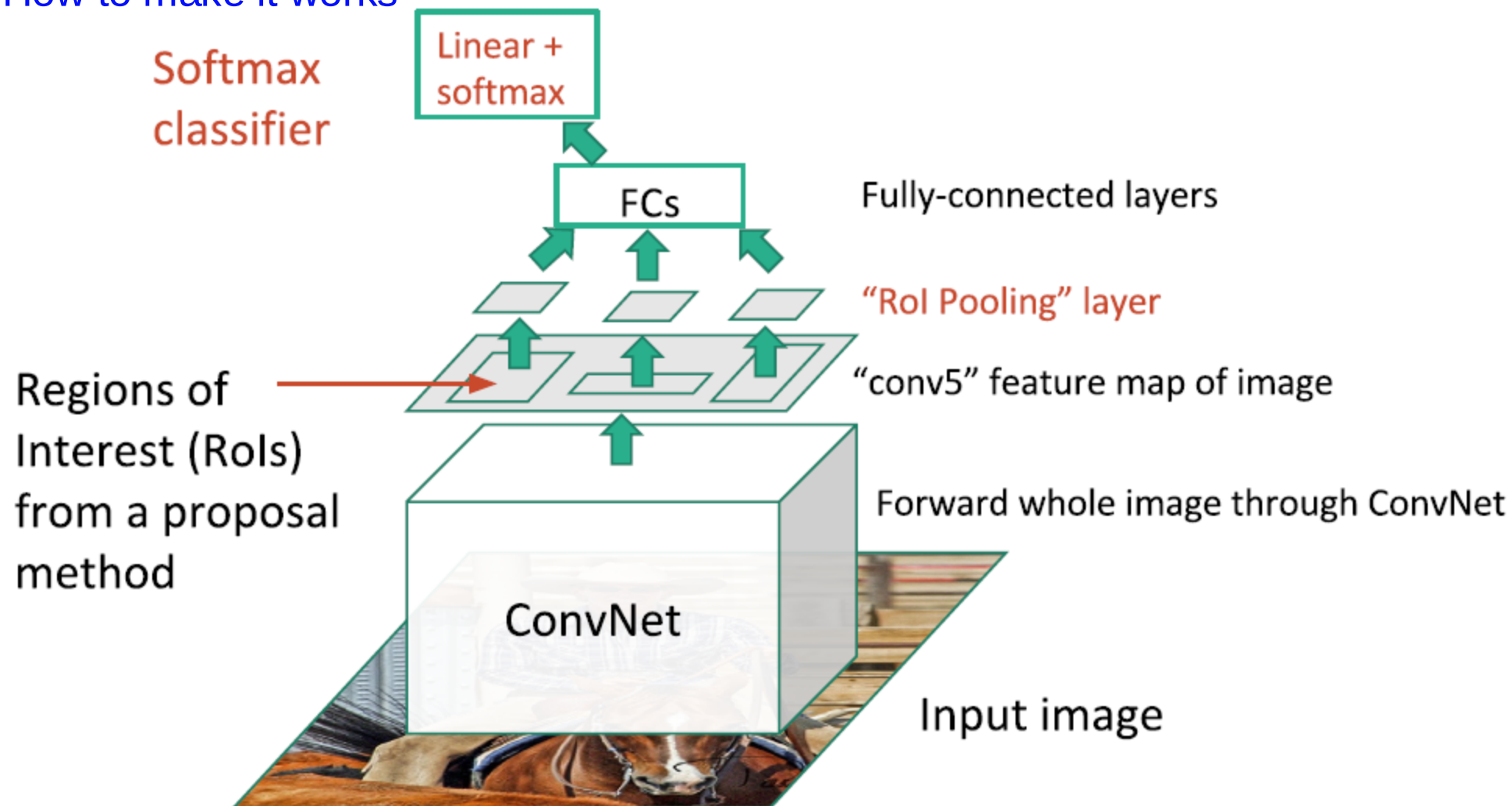
1. How to make it works



3.1 Fast R-CNN

□ Main idea

1. How to make it works

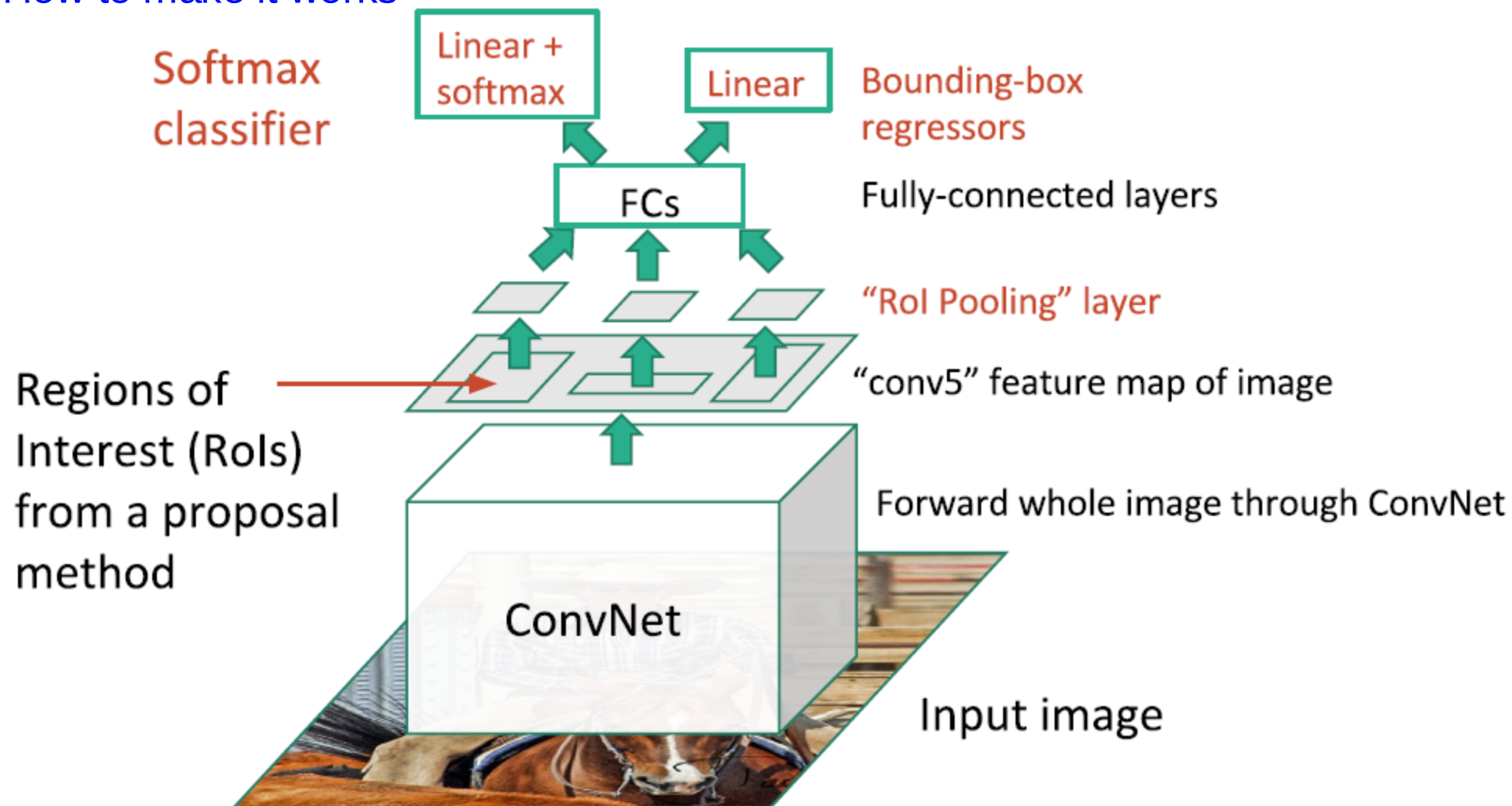


Girshick, "Fast R-CNN", ICCV 2015.

3.1 Fast R-CNN

□ Main idea

1. How to make it works



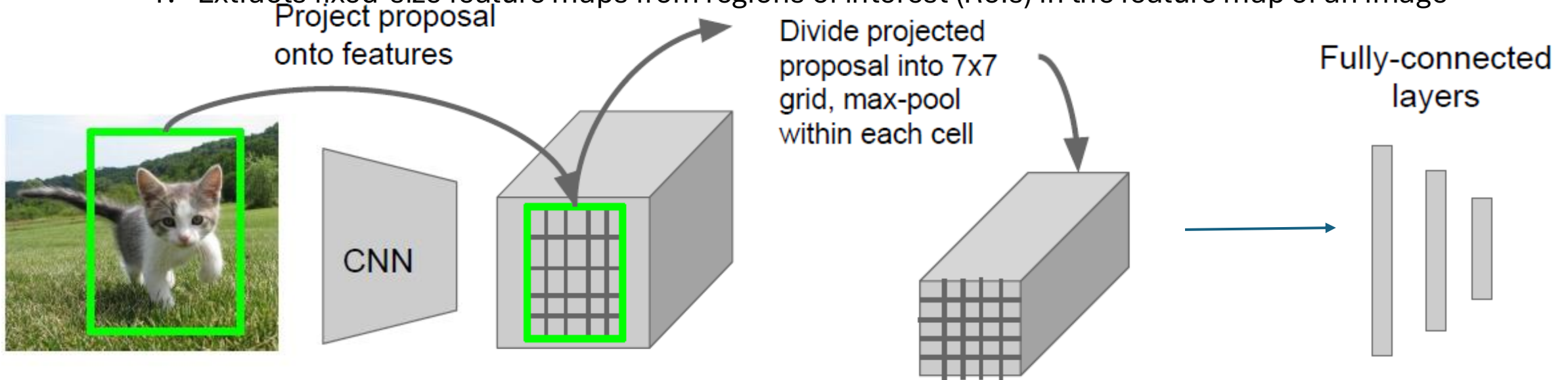
Girshick, "Fast R-CNN", ICCV 2015.

3.1 Fast R-CNN : RoI- Pooling

□ Main idea

RoI (Region of Interest) Pooling is a key operation in Faster R-CNN

1. Extracts fixed-size feature maps from regions of interest (Rois) in the feature map of an image



Input:

- A **feature map** (output of the CNN backbone).
- A set of **RoIs** (bounding boxes) proposed by the Region Proposal Network (RPN).
- Each RoI is represented by coordinates on the feature map.

Pooling (Max Pooling):

- For each grid cell, select the corresponding region in the feature map.
- Apply max pooling to reduce the variable-sized feature values to a single value for that cell.

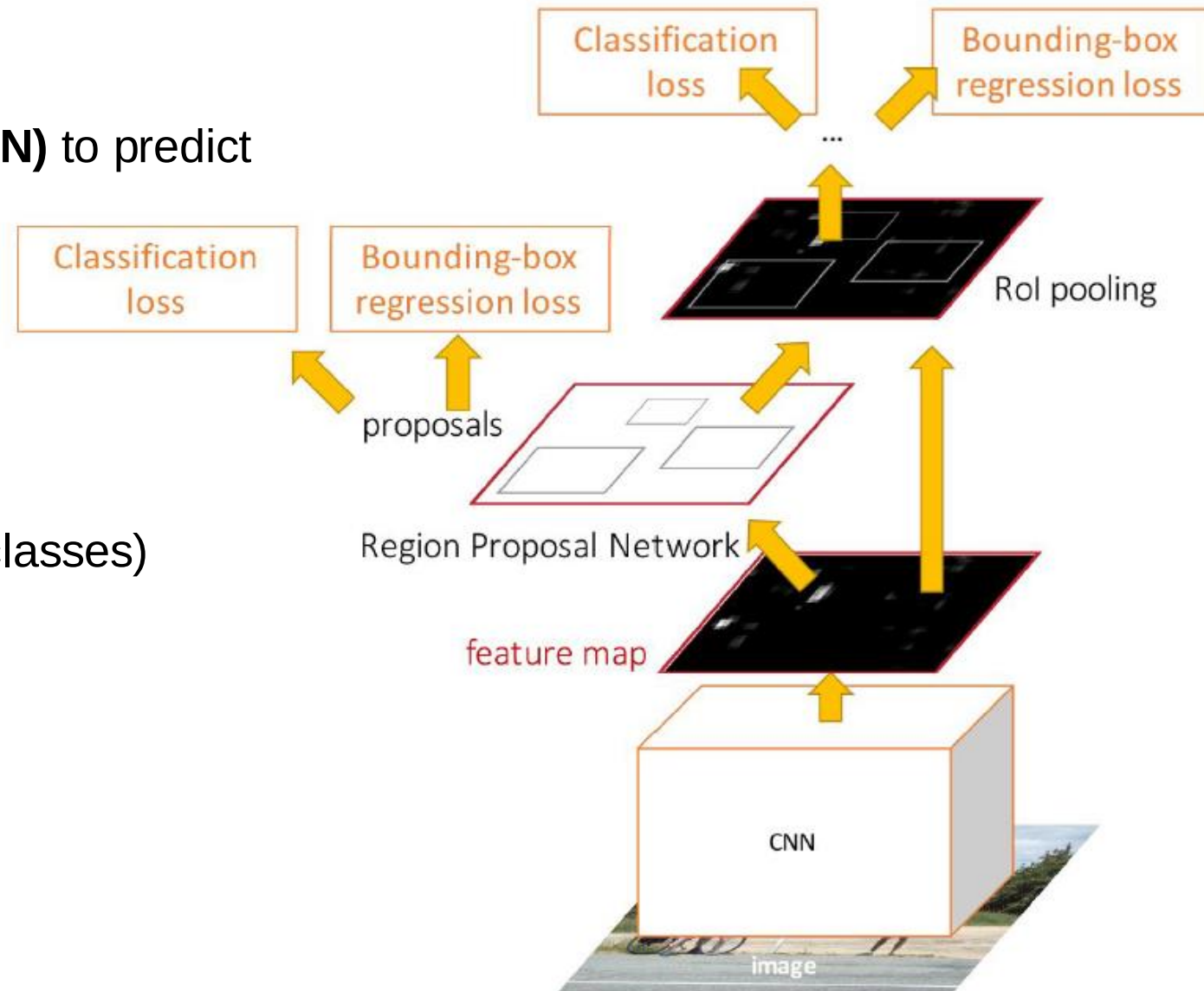
3.2 Faster R-CNN

□ Main idea

Insert **Region Proposal Network (RPN)** to predict proposals from features

Jointly train with 4 losses:

1. RPN classify object / not object
2. RPN regress box coordinates
3. Final classification score (object classes)
4. Final box coordinates



Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015

3.3 Detection without Proposals: YOLO / SSD

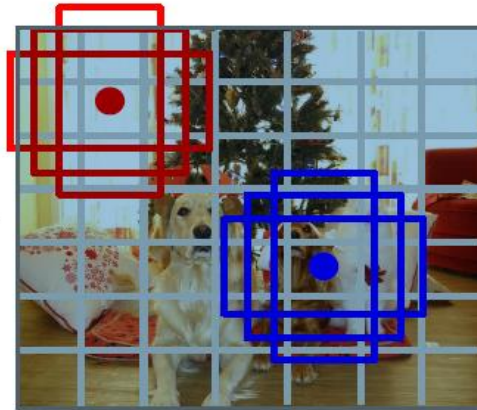
□ Main idea

YOLO : You Only Look Once / SSD: Single-Shot MultiBox Detector

Go from input image to tensor of scores with one big convolutional network!



Input image $3 \times H \times W$



Divide image into grid 7×7

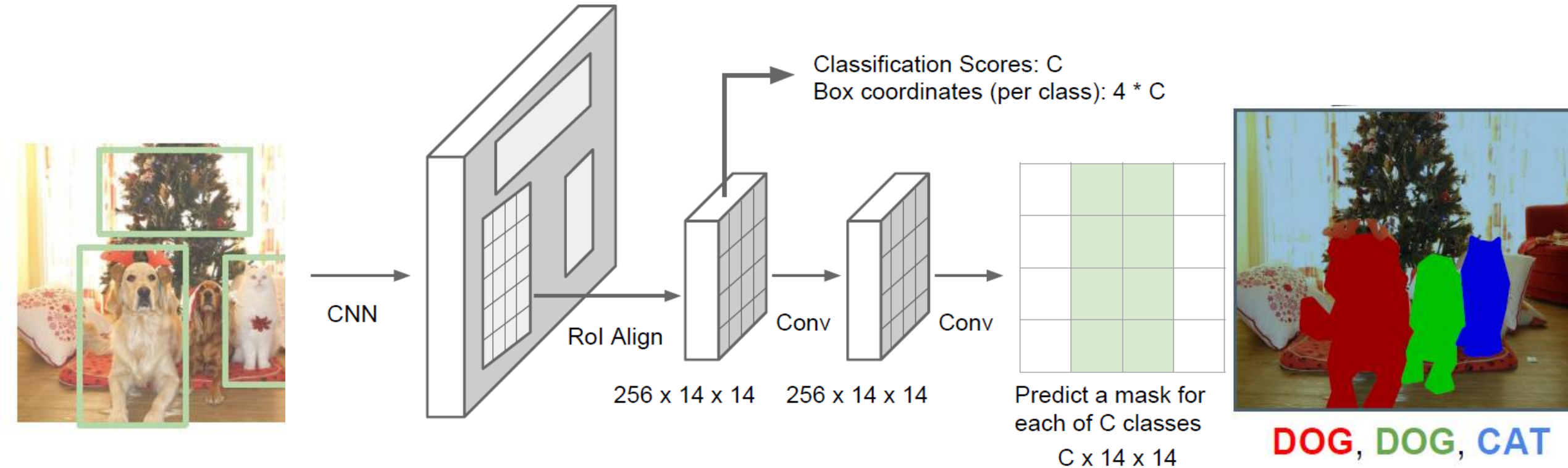
Within each grid cell:

- Regress from each of the B base boxes to a final box with 5 numbers: (dx, dy, dh, dw, confidence)
- Predict scores for each of C classes (including background as a class)

4. Instance Segmentation

□ Main idea

Mask R-CNN



He et al, "Mask R-CNN", arXiv 2017